

1) V.m / Variància / σ / Quantil / Ho.9

$$a - E[X] = 0 \times 0,05 + 1 \times 0,1 + 2 \times 0,2 + 3 \times 0,15 + 4 \times 0,3 + 5 \times 0,2$$

$$= 3,15$$

$$\text{Var}[X] = (0-3,15)^2 \times 0,05 + (1-3,15)^2 \times 0,1 + (2-3,15)^2 \times 0,2$$

$$+ (3-3,15)^2 \times 0,15 + (4-3,15)^2 \times 0,3 + (5-3,15)^2 \times 0,2$$

$$= 0,4961 + 0,4623 + 0,2645 + 0,00375$$

$$+ 0,2168 + 0,6845$$

$$= 2,1280$$

$$\sigma = \sqrt{2,1280}$$

$$F(c) = \begin{cases} 0 & \text{re } x < 0 \\ 0,05 & \text{re } 0 \leq x < 1 \\ 0,15 & \text{re } 1 \leq x < 2 \\ 0,35 & \text{re } 2 \leq x < 3 \\ 0,5 & \text{re } 3 \leq x < 4 \\ 0,8 & \text{re } 4 \leq x < 5 \\ 1 & \text{re } x \geq 5 \end{cases}$$

$$X_{0,25} = 2$$

$$X_{0,5} = 3$$

$$X_{0,75} = 4$$

$$X_{0,9} = 5$$

b- Exercício 2

$$M_x = \int_1^3 x \cdot \frac{x}{4} dx = \frac{1}{4} \int_1^3 x^2 dx = \frac{1}{4} \cdot \left[\frac{x^3}{3} \right]_1^3$$

$$= \frac{1}{4} \cdot \left[\frac{3^3}{3} - \frac{1^3}{3} \right] = \frac{1}{4} \cdot \left(9 - \frac{1}{3} \right) = \frac{1}{4} \cdot \frac{26}{3} = \frac{26}{12} = \frac{13}{6}$$

$$\text{Var}[X] = \int_1^3 x^2 \cdot \frac{x}{4} dx - \left(\frac{13}{6} \right)^2$$

$$= \frac{1}{4} \cdot \int_1^3 x^3 dx - \left(\frac{13}{6}\right)^2$$

$$= \frac{1}{4} \cdot \left[\frac{x^4}{4}\right]_1^3 - \left(\frac{13}{6}\right)^2$$

$$= \frac{1}{4} \cdot \left[\frac{81}{4} - \frac{1}{4}\right] - \left(\frac{13}{6}\right)^2 = 5 - \frac{169}{36} = \frac{11}{36}$$

$$\sigma_x = \frac{\sqrt{11}}{6}$$

$$F(x) = \int_1^x \frac{t}{4} dt = \frac{1}{4} \int_1^x t dt$$

$$= \frac{1}{4} \cdot \left(\frac{x^2}{2} - \frac{1}{2}\right) = \frac{x^2 - 1}{8}$$

$$\underline{K_{0.25}}$$

$$F(x) = 0,25$$

$$\frac{x^2 - 1}{8} = 0,25$$

$$\Leftrightarrow x^2 = 3$$

$$\Leftrightarrow x = \sqrt{3}, \quad x > 0$$

$$\underline{K_{0.5}}$$

$$\frac{x^2 - 1}{8} = 0,5$$

$$\Leftrightarrow x = \sqrt{5}$$

$$\underline{K_{0.75}}$$

$$\frac{x^2 - 1}{8} = 0,75$$

$$\Leftrightarrow x = \sqrt{7}$$

$$\underline{K_{0.9}}$$

$$\frac{x^2 - 1}{8} = 0,9$$

$$\Leftrightarrow x = \sqrt{8.2}$$

Exercício 3

$$f(u) = \begin{cases} u+1 & \text{se } -1 < u \leq 0 \\ -u+1 & \text{se } 0 < u \leq 1 \\ 0 & \text{c.c.} \end{cases}$$

$$E[X] = \int_{-1}^1 u f(u) du = \int_{-1}^0 u(u+1) du + \int_0^1 u(-u+1) du$$

$$= \int_{-1}^0 (u^2 + u) du + \int_0^1 (-u^2 + u) du$$

$$= \left[\frac{u^3}{3} + \frac{u^2}{2} \right]_{-1}^0 + \left[-\frac{u^3}{3} + \frac{u^2}{2} \right]_0^1$$

$$= -\frac{1}{3} + \frac{1}{2} + \left(-\left(-\frac{1}{3} + \frac{1}{2}\right) \right)$$

$$= \frac{1}{6} - \frac{1}{6} = 0$$

$$Var[X] = \int_{-1}^1 u^2 f(u) du - 0 = \int_{-1}^0 u^2(u+1) du + \int_0^1 u^2(-u+1) du$$

$$= \int_{-1}^0 (u^3 + u^2) du + \int_0^1 (-u^3 + u^2) du$$

$$= \left[\frac{u^4}{4} + \frac{u^3}{3} \right]_{-1}^0 + \left[-\frac{u^4}{4} + \frac{u^3}{3} \right]_0^1$$

$$= \left[0 - \left(\frac{1}{4} - \frac{1}{3} \right) \right] + \left(-\frac{1}{4} + \frac{1}{3} - 0 \right)$$

$$= \frac{1}{12} + \frac{1}{12} = \frac{1}{6}$$

$$\sigma_x = \frac{\sqrt{6}}{6}$$

Para $-1 \leq u \leq 0$

$$\int_{-1}^u (u+1) du = \left[\frac{u^2}{2} + u \right]_{-1}^u = \frac{u^2}{2} + u + \frac{1}{2}$$

Para $0 \leq u \leq 1$

$$\frac{1}{2} + \int_0^u (-u+1) du = \left[-\frac{u^2}{2} + u \right]_0^u + \frac{1}{2} = -\frac{u^2}{2} + u + \frac{1}{2}$$

↓

de $-1 \leq u \leq 0$ do
acumulamos $1/2$
de probabilidade

$$X_{0.25}$$

Como $0,25 < 0,5$, vai estar na primeira metade

$$0,25 = \frac{u^2}{2} + u + \frac{1}{2}$$

$$\Leftrightarrow 0 = \frac{u^2}{2} + u + \frac{1}{4}$$

$$\Leftrightarrow u = \sqrt{2} - 1, \quad u > 0$$

$$X_{0.5} = 0$$

$$X_{0.75}$$

$$0,75 = -\frac{u^2}{2} + u + \frac{1}{2}$$

$$\Leftrightarrow -\frac{u^2}{2} + u - \frac{1}{4} = 0$$

$$\Leftrightarrow u = 1 - \frac{\sqrt{2}}{2}, \quad 0 \leq u \leq 1$$

$$2) \text{ a- } E[x] = \int_0^1 12u^2(1-u) \cdot u \, du$$

$$= 12 \int_0^1 (u^3 - u^4) \, du$$

$$= 12 \left[\frac{u^4}{4} - \frac{u^5}{5} \right]_0^1$$

$$= 12 \times \left[\frac{1}{4} - \frac{1}{5} \right] = \frac{12}{20} = \frac{6}{10} = \frac{3}{5}$$

$$\text{Var}[x] = \int_0^1 12u^2(1-u) \cdot u^2 \cdot du - \left(\frac{3}{5}\right)^2$$

$$= 12 \int_0^1 (u^4 - u^5) \, du - \frac{9}{25}$$

$$= 12 \times \left[\frac{u^5}{5} - \frac{u^6}{6} \right]_0^1 - \frac{9}{25}$$

$$= 12 \times \left[\frac{1}{5} - \frac{1}{6} \right] - \frac{9}{25} = \frac{1}{25}$$

Para $0 < u < 1$

$$F(u) = \int_0^u 12u^2(1-u) \, du = 12 \int_0^u (u^2 - u^3) \, du$$

$$= 12 \cdot \left[\frac{u^3}{3} - \frac{u^4}{4} \right]_0^u$$

$$= 12 \cdot \left(\frac{u^3}{3} - \frac{u^4}{4} \right)$$

$$= 4u^3 - 3u^4$$

$$F_x(u) = \begin{cases} 4u^3 - 3u^4 & \text{se } 0 < u < 1 \\ 0 & \text{se c.c.} \end{cases}$$

$$b- i) P(x < 1/3) = 4 \times \left(\frac{1}{3}\right)^3 - 3 \cdot \left(\frac{1}{3}\right)^4$$

$$= \frac{1}{9}$$

$$P(1/3 \leq x < 2/3) = F(2/3) - F(1/3)$$

$$= \left(4 \times \left(\frac{2}{3}\right)^3 - 3 \times \left(\frac{2}{3}\right)^4\right) - \frac{1}{9}$$

$$= \frac{13}{27}$$

$$P(x \geq 2/3) = 1 - F(2/3)$$

$$= 1 - \frac{16}{27} = \frac{11}{27}$$

$$L: \begin{cases} 8 & 13 & 18 \\ \frac{1}{9} & \frac{13}{27} & \frac{11}{27} \end{cases}$$

$$ii) E[x] = 8 \times \frac{1}{9} + 13 \times \frac{13}{27} + 18 \times \frac{11}{27}$$

$$= \frac{8}{9} + \frac{169}{27} + \frac{198}{27}$$

$$= \frac{391}{27}$$

c) Distribuição Binomial

$$P(x=k) = \binom{5}{k} \cdot \left(\frac{1}{9}\right)^k \cdot \left(\frac{8}{9}\right)^{5-k}$$

$$y: \begin{cases} 0 & 1 & 2 & 3 & 4 & 5 \\ \left(\frac{8}{9}\right)^5 & 5 \cdot \frac{8^4}{9^5} & 10 \times \frac{8^3}{9^5} & 10 \times \frac{8^2}{9^5} & 5 \times \frac{8}{9^5} & \frac{1}{9^5} \end{cases}$$